7 The setup in Fig. 7.1 is used to investigate the photoelectric effect.

Monochromatic electromagnetic radiation of wavelength λ is incident on a metal emitter plate E. The emitted photoelectrons from E reach the collector plate C and are detected by the ammeter as a current. The potential difference across E and C can be adjusted via the variable voltage source.

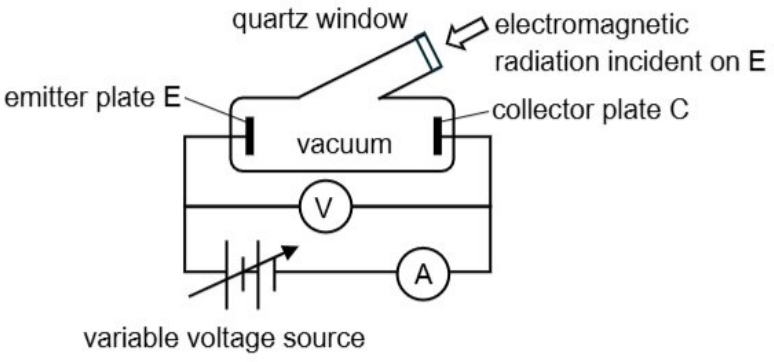


Fig. 7.1

The wavelength of the light λ is varied and the minimum potential difference V_s across E and C that results in no current being detected in the ammeter is recorded. The graph in Fig. 7.2 shows the variation of V_s with λ.

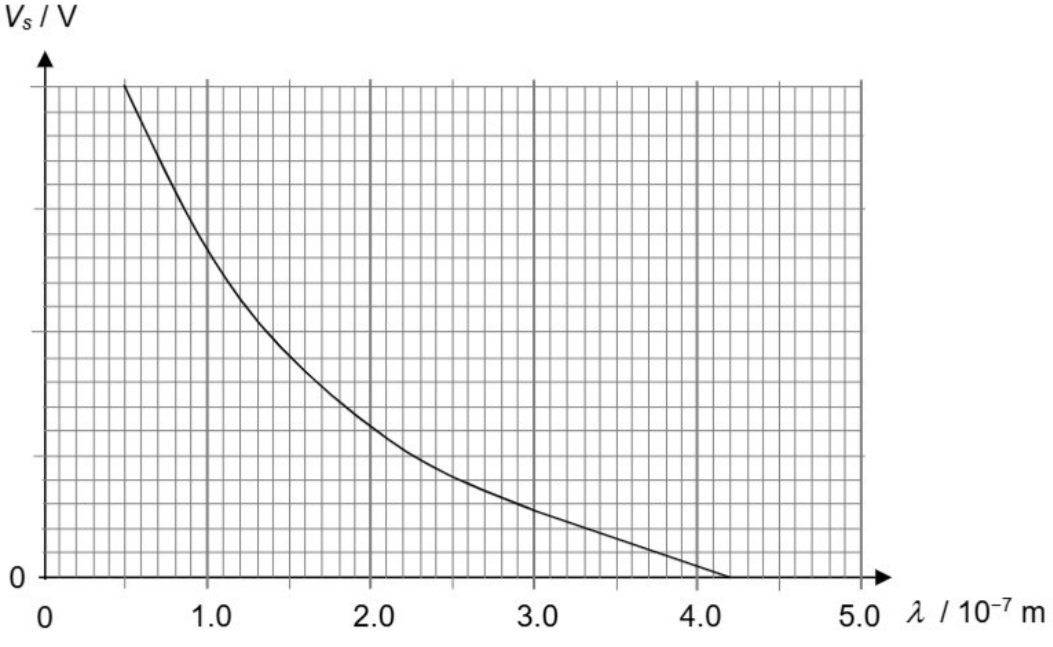


Fig. 7.2

(a) State the significance of the horizontal intercept of the graph.

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..... [1]

(b) **(i)** Calculate the minimum photon energy required to allow an electron to overcome the surface energy barrier and be emitted from the emitter plate.

energy = ... J [2]

(ii) Determine the maximum kinetic energy of the emitted electron when the wavelength used is 2.0×10^{-7} m.

kinetic energy = ... J [2]

(c) Explain how the existence of a threshold frequency in the photoelectric effect provides evidence for the particulate nature of electromagnetic radiation.

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..... [3]